

0.1 定积分

0.1.1 建立积分递推

例题 0.1 计算 $\int_0^{\frac{\pi}{2}} \cos^n x \sin(nx) dx, n \in \mathbb{N}$.

证明 Show/Hide Proof

利用分部积分和和差化积公式可得

$$\begin{aligned} I_n &= \int_0^{\frac{\pi}{2}} \cos^n x \sin(nx) dx = \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cdot \cos x \sin(nx) dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cdot [\sin((n+1)x) + \sin((n-1)x)] dx \\ &= \frac{I_{n-1}}{2} + \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^{n-1} x [\sin(nx) \cos x + \cos(nx) \sin x] dx \\ &= \frac{I_{n-1}}{2} + \frac{I_n}{2} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^{n-1} x \cos(nx) d \cos x \\ &= \frac{I_{n-1} + I_n}{2} - \frac{1}{2n} \int_0^{\frac{\pi}{2}} \cos(nx) d \cos^n x \\ &= \frac{I_{n-1} + I_n}{2} + \frac{1}{2n} - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^n x \sin(nx) dx \\ &= \frac{I_{n-1} + I_n}{2} + \frac{1}{2n} - \frac{I_n}{2} \\ &= \frac{I_{n-1}}{2} + \frac{1}{2n}. \end{aligned}$$

故 $I_n = \frac{I_{n-1}}{2} + \frac{1}{2n}$, 则两边同乘 2^n (强行裂项)

$$2^n I_n = 2^{n-1} I_{n-1} + \frac{2^{n-1}}{n}, n = 1, 2, \dots$$

又注意到 $I_0 = 0$, 从而

$$2^n I_n = 0 + \sum_{k=1}^n \frac{2^{k-1}}{k} \Rightarrow I_n = \frac{1}{2^n} \sum_{k=1}^n \frac{2^{k-1}}{k}.$$

□

命题 0.1

设 $n \in \mathbb{N}$, 证明:

- (1) $\int_0^{\pi} \frac{\sin(nx)}{\sin x} dx = \begin{cases} 0, & n \text{ 为偶数} \\ \pi, & n \text{ 为奇数} \end{cases}.$
- (2) $\int_0^{\pi} \frac{\sin^2(nx)}{\sin^2 x} dx = n\pi.$
- (3) $\int_0^{\pi} \frac{\sin^2(nx)}{\sin x} dx = \sum_{k=1}^n \frac{2}{2k-1}.$

▲

 **笔记** Show/Hide Note

提示: $\sin^2 x - \sin^2 y = \sin(x-y) \sin(x+y)$ (证明见命题??).

证明 Show/Hide Proof

(1) 记 $I_n = \int_0^{\pi} \frac{\sin(nx)}{\sin x} dx$, 则

$$I_{n+2} - I_n = \int_0^{\pi} \frac{\sin((n+2)x) - \sin(nx)}{\sin x} dx = \int_0^{\pi} \frac{2 \cos((n+1)x) \sin x}{\sin x} dx = 2 \int_0^{\pi} \cos((n+1)x) dx = 0.$$

于是

$$\int_0^{\pi} \frac{\sin(nx)}{\sin x} dx = I_n = I_{n-2} = \cdots = \begin{cases} I_0, & n \text{ 为偶数} \\ I_1, & n \text{ 为奇数} \end{cases} = \begin{cases} 0, & n \text{ 为偶数} \\ \pi, & n \text{ 为奇数} \end{cases}.$$

(2) 记 $I_n = \int_0^{\pi} \frac{\sin^2(nx)}{\sin^2 x} dx$, 则

$$\begin{aligned} I_{n+1} - I_n &= \int_0^{\pi} \frac{\sin^2((n+1)x) - \sin^2(nx)}{\sin^2 x} dx = \int_0^{\pi} \frac{\sin x \cdot \sin((2n+1)x)}{\sin^2 x} dx \\ &= \int_0^{\pi} \frac{\sin((2n+1)x)}{\sin x} dx \stackrel{\text{命题 0.1(1)}}{=} \pi. \end{aligned} \quad (1)$$

于是

$$\int_0^{\pi} \frac{\sin^2(nx)}{\sin^2 x} dx = I_n = \pi + I_{n-1} = \cdots = (n-1)\pi + I_1 = n\pi.$$

(3) 记 $I_n = \int_0^{\pi} \frac{\sin^2(nx)}{\sin x} dx$, 则

$$\begin{aligned} I_{n+1} - I_n &= \int_0^{\pi} \frac{\sin^2((n+1)x) - \sin^2(nx)}{\sin x} dx = \int_0^{\pi} \frac{\sin x \cdot \sin((2n+1)x)}{\sin x} dx \\ &= \int_0^{\pi} \sin((2n+1)x) dx = \frac{1}{2n+1} \cos((2n+1)x) \Big|_{\pi}^0 = \frac{2}{2n+1}. \end{aligned} \quad (2)$$

于是

$$\int_0^{\pi} \frac{\sin^2(nx)}{\sin x} dx = I_n = \frac{2}{2n-1} + I_{n-1} = \cdots = \sum_{k=1}^{n-1} \frac{2}{2k+1} + I_1 = \sum_{k=0}^{n-1} \frac{2}{2k+1} = \sum_{k=1}^n \frac{2}{2k-1}.$$

□

例题 0.2 设 $a > 1$, 计算积分 $\int_0^{\frac{\pi}{2}} \ln(a^2 - \cos^2 x) dx$.

注 很多情况下不需求出被积函数的原函数, 只需充分利用换元、分部积分以及被积函数的性质, 即可求出积分的值. 见下述解法二.

解 [Show/Hide Solution](#)

解法一: 设 $a_0 = a > 1$. 构造数列如下:

$$a_{n+1} = 2a_n^2 - 1 \quad (n = 0, 1, \cdots),$$

则由例题??可知, 存在 $x_0 > 0$ 使得

$$a_0 = \operatorname{ch}(x_0), \quad a_n = \operatorname{ch}(2^n x_0),$$

其中 $\operatorname{ch}(x) = \frac{1}{2}(e^x + e^{-x})$. 可以解得

$$x_0 = \ln \left(a_0 + \sqrt{a_0^2 - 1} \right). \quad (3)$$

故

$$a_n = \frac{e^{2^n x_0} + e^{-2^n x_0}}{2}.$$

设

$$I_n = \int_0^{\pi} \ln(a_n - \cos x) dx,$$

则

$$\begin{aligned} I_0 &= \int_0^{\pi} \ln(a_0 - \cos x) dx = \int_0^{\frac{\pi}{2}} \ln(a_0 - \cos x) dx + \int_{\frac{\pi}{2}}^{\pi} \ln(a_0 - \cos x) dx \\ &= \int_0^{\frac{\pi}{2}} \ln(a_0 - \cos x) dx + \int_0^{\frac{\pi}{2}} \ln(a_0 + \cos x) dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\frac{\pi}{2}} \ln(a_0^2 - \cos^2 x) dx = \int_0^{\frac{\pi}{2}} \ln\left(a_0^2 - \frac{1 + \cos 2x}{2}\right) dx \\
&= \int_0^{\frac{\pi}{2}} \ln\left(\frac{a_1 - \cos 2x}{2}\right) dx = \frac{1}{2} \int_0^{\pi} \ln\left(\frac{a_1 - \cos x}{2}\right) dx = \frac{1}{2} I_1 - \frac{\pi}{2} \ln 2.
\end{aligned}$$

同理, 有

$$I_n = \frac{1}{2} I_{n+1} - \frac{\pi}{2} \ln 2. \quad (4)$$

由此递推公式, 可得

$$I_0 = \frac{1}{2^n} I_n - \left(1 + \frac{1}{2} + \cdots + \frac{1}{2^{n-1}}\right) \frac{\pi}{2} \ln 2. \quad (5)$$

因为

$$\begin{aligned}
I_n &= \int_0^{\pi} \ln(a_n - \cos x) dx = \int_0^{\pi} \ln\left(\frac{e^{2^n x_0} + e^{-2^n x_0}}{2} - \cos x\right) dx \\
&= 2^n x_0 \pi + \int_0^{\pi} \ln\left(\frac{1 + e^{-2^{n+1} x_0}}{2} - e^{-2^n x_0} \cos x\right) dx,
\end{aligned}$$

所以

$$\frac{1}{2^n} I_n \rightarrow x_0 \pi \quad (n \rightarrow +\infty).$$

故从式 (5) 可得

$$I_0 = x_0 \pi - \pi \ln 2 = \pi \ln \left(\frac{a_0 + \sqrt{a_0^2 - 1}}{2} \right),$$

即所求的积分为

$$\int_0^{\frac{\pi}{2}} \ln(a^2 - \sin^2 x) dx = \int_0^{\frac{\pi}{2}} \ln(a^2 - \cos^2 x) dx = \pi \ln \left(\frac{a + \sqrt{a^2 - 1}}{2} \right).$$

解法二: 我们有

$$F(a) = \int_0^{\frac{\pi}{2}} \ln(a^2 - \cos^2 x) dx = \int_0^{\pi} \ln(a - \cos x) dx.$$

由定理??, 关于 a 求导得到

$$F'(a) = \int_0^{\pi} \frac{1}{a - \cos x} dx \stackrel{\text{万能公式}}{=} \int_0^{+\infty} \frac{2}{a(1+t^2) - (1-t^2)} dt = \frac{\pi}{\sqrt{a^2 - 1}}, \quad a > 1.$$

因此

$$F(a) = \int_1^a F'(a) da = \pi \ln(a + \sqrt{a^2 - 1}) + C, \quad a > 1.$$

结合

$$F(1) = 2 \int_0^{\frac{\pi}{2}} \ln \sin x dx = -\pi \ln 2.$$

可得

$$\int_0^{\frac{\pi}{2}} \ln(a^2 - \cos^2 x) dx = \pi \ln \left(\frac{a + \sqrt{a^2 - 1}}{2} \right), \quad a > 1.$$

□

命题 0.2 (周期函数在任意一个周期上的定积分相同)

设 f 是 $\mathbb{R}$ 上的可积函数且周期为 $T > 0$, 则对 $\forall a \in \mathbb{R}$, 都有

$$\int_a^{a+T} f(x) dx = \int_0^T f(x) dx.$$

◆

证明 Show/Hide Proof

由条件可得

$$\begin{aligned}\int_a^{a+T} f(x)dx &= \int_a^{(\lfloor \frac{a}{T} \rfloor + 1)T} f(x)dx + \int_{(\lfloor \frac{a}{T} \rfloor + 1)T}^{a+T} f(x)dx = \int_a^{(\lfloor \frac{a}{T} \rfloor + 1)T} f(x)dx + \int_{\lfloor \frac{a}{T} \rfloor T}^a f(x+T)dx \\ &= \int_a^{(\lfloor \frac{a}{T} \rfloor + 1)T} f(x)dx + \int_{\lfloor \frac{a}{T} \rfloor T}^a f(x)dx = \int_{\lfloor \frac{a}{T} \rfloor T}^{(\lfloor \frac{a}{T} \rfloor + 1)T} f(x)dx \\ &= \int_0^T f\left(x + \lfloor \frac{a}{T} \rfloor T\right) dx = \int_0^T f(x)dx.\end{aligned}$$

□

定理 0.1 (点火公式)

1. 记 $I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx = \int_0^{\frac{\pi}{2}} \cos^n x dx, \forall n \in \mathbb{N}$, 则

$$I_n = \frac{n-1}{n} I_{n-2}, \quad \forall n \geq 2.$$

从而

$$I_n = \begin{cases} \frac{(n-1)!!}{n!!} I_0 = \frac{(n-1)!!}{n!!} \cdot \frac{\pi}{2}, & n \text{ 为偶数} \\ \frac{(n-1)!!}{n!!} I_1 = \frac{(n-1)!!}{n!!}, & n \text{ 为奇数} \end{cases}. \quad (6)$$

2. 记 $J(m, n) = \int_0^{\frac{\pi}{2}} \sin^m x \cos^n x dx, \forall n, m \in \mathbb{N}$, 则

$$J(m, n) = \frac{m-1}{m+n} J(m-2, n), \quad \forall n, m \geq 2.$$

$$J(m, n) = \frac{n-1}{m+n} J(m, n-2), \quad \forall n, m \geq 2.$$

从而

$$J(m, n) = \begin{cases} \frac{(m-1)!!(n-1)!!}{(m+n)!!}, & m, n \text{ 不全为偶数} \\ \frac{(m-1)!!(n-1)!!}{(m+n)!!} \cdot \frac{\pi}{2}, & m, n \text{ 全为偶数} \end{cases}. \quad (7)$$

♡

0.1.2 区间再现

定理 0.2 (区间再现恒等式)

当下述积分有意义时, 我们有

1.

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx = \frac{1}{2} \int_a^b [f(x) + f(a+b-x)]dx = \int_a^{\frac{a+b}{2}} [f(x) + f(a+b-x)]dx.$$

2.

$$\int_0^\infty f(x)dx = \int_0^1 f(x)dx + \int_1^\infty f(x)dx = \int_0^1 \left[f(x) + \frac{f(\frac{1}{x})}{x^2} \right] dx.$$

♡



笔记 Show/Hide Note

注意: 倒代换具有将 $[0, 1]$ 转化为 $[1, +\infty)$ 的功能.

证明 Show/Hide Proof

证明是显然的.(第 1 问中最后一个等号是由轴对称得到的)

□

命题 0.3

证明

1. $\int_0^{\frac{\pi}{2}} \ln \sin x dx = -\frac{\pi}{2} \ln 2.$
2. $\int_0^{\frac{\pi}{2}} \ln \cos x dx = -\frac{\pi}{2} \ln 2.$
3. $\int_0^1 \frac{\ln(1+x)}{1+x^2} dx = \frac{\pi}{8} \ln 2.$

证明 Show/Hide Proof

1.

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \ln \sin x dx = \int_0^{\frac{\pi}{4}} \left[\ln \sin x + \ln \sin \left(\frac{\pi}{2} - x \right) \right] dx \\
 &= \int_0^{\frac{\pi}{4}} \ln (\cos x \sin x) dx = \int_0^{\frac{\pi}{4}} \ln \frac{1}{2} dx + \int_0^{\frac{\pi}{4}} \ln (\sin 2x) dx \\
 &= -\frac{\pi}{4} \ln 2 + \frac{1}{2} \int_0^{\frac{\pi}{2}} \ln \sin x dx = -\frac{\pi}{4} \ln 2 + \frac{1}{2} I \\
 \implies I &= \int_0^{\frac{\pi}{2}} \ln \cos x dx = -\frac{\pi}{2} \ln 2.
 \end{aligned}$$

2.

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \ln \cos x dx = \int_0^{\frac{\pi}{2}} \ln \sin x dx = \int_0^{\frac{\pi}{4}} \left[\ln \sin x + \ln \sin \left(\frac{\pi}{2} - x \right) \right] dx \\
 &= \int_0^{\frac{\pi}{4}} \ln (\cos x \sin x) dx = \int_0^{\frac{\pi}{4}} \ln \frac{1}{2} dx + \int_0^{\frac{\pi}{4}} \ln (\sin 2x) dx \\
 &= -\frac{\pi}{4} \ln 2 + \frac{1}{2} \int_0^{\frac{\pi}{2}} \ln \sin x dx = -\frac{\pi}{4} \ln 2 + \frac{1}{2} I \\
 \implies I &= \int_0^{\frac{\pi}{2}} \ln \cos x dx = -\frac{\pi}{2} \ln 2.
 \end{aligned}$$

3. 解法一:

$$\begin{aligned}
 \int_0^1 \frac{\ln(1+x)}{1+x^2} dx &\stackrel{x=\tan \theta}{=} \int_0^{\frac{\pi}{4}} \frac{\ln(1+\tan \theta)}{1+\tan^2 \theta} d \tan \theta = \int_0^{\frac{\pi}{4}} \frac{\sec^2 \theta \cdot \ln(1+\tan \theta)}{\sec^2 \theta} d\theta \\
 &= \int_0^{\frac{\pi}{4}} \ln(1+\tan \theta) d\theta = \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan \theta) + \ln \left(1 + \tan \left(\frac{\pi}{4} - \theta \right) \right) \right] d\theta \\
 &= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan \theta) + \ln \left(1 + \frac{1-\tan \theta}{1+\tan \theta} \right) \right] d\theta \\
 &= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan \theta) + \ln \frac{2}{1+\tan \theta} \right] d\theta \\
 &= \int_0^{\frac{\pi}{8}} \ln 2 d\theta = \frac{\pi}{8} \ln 2.
 \end{aligned}$$

解法二:考虑含参量积分

$$\varphi(\alpha) = \int_0^1 \frac{\ln(1+\alpha x)}{1+x^2} dx, \quad \alpha \in [0, 1].$$

显然 $\varphi(0) = 0, \varphi(1) = I$, 且函数 $\frac{\ln(1+\alpha x)}{1+x^2}$ 在 $R = [0, 1] \times [0, 1]$ 上满足定理?? 的条件, 于是

$$\varphi'(\alpha) = \int_0^1 \frac{x}{(1+x^2)(1+\alpha x)} dx.$$

因为

$$\frac{x}{(1+x^2)(1+\alpha x)} = \frac{1}{1+\alpha^2} \left(\frac{\alpha+x}{1+x^2} - \frac{\alpha}{1+\alpha x} \right),$$

所以

$$\begin{aligned}\varphi'(\alpha) &= \frac{1}{1+\alpha^2} \left(\int_0^1 \frac{\alpha}{1+x^2} dx + \int_0^1 \frac{x}{1+x^2} dx - \int_0^1 \frac{\alpha}{1+\alpha x} dx \right) \\ &= \frac{1}{1+\alpha^2} \left[\alpha \arctan x \Big|_0^1 + \frac{1}{2} \ln(1+x^2) \Big|_0^1 - \ln(1+\alpha x) \Big|_0^1 \right] \\ &= \frac{1}{1+\alpha^2} \left[\alpha \cdot \frac{\pi}{4} + \frac{1}{2} \ln 2 - \ln(1+\alpha) \right].\end{aligned}$$

因此

$$\begin{aligned}\int_0^1 \varphi'(\alpha) d\alpha &= \int_0^1 \frac{1}{1+\alpha^2} \left[\frac{\pi}{4} \alpha + \frac{1}{2} \ln 2 - \ln(1+\alpha) \right] d\alpha \\ &= \frac{\pi}{8} \ln(1+\alpha^2) \Big|_0^1 + \frac{1}{2} \ln 2 \arctan \alpha \Big|_0^1 - \varphi(1) \\ &= \frac{\pi}{8} \ln 2 + \frac{\pi}{8} \ln 2 - \varphi(1) \\ &= \frac{\pi}{4} \ln 2 - \varphi(1).\end{aligned}$$

另一方面,

$$\int_0^1 \varphi'(\alpha) d\alpha = \varphi(1) - \varphi(0) = \varphi(1),$$

所以 $I = \varphi(1) = \frac{\pi}{8} \ln 2$.

□

例题 0.3 计算

1. $\int_0^\infty \frac{\ln x}{x^2+a^2} dx, a > 0.$
2. $\int_0^\infty \frac{\ln x}{x^2+x+1} dx.$
3. $\int_0^1 \frac{\ln x}{\sqrt{x-x^2}} dx.$

解 Show/Hide Solution

1. 注意到

$$\int_0^{+\infty} \frac{\ln x}{x^2+a^2} dx \stackrel{x=at}{=} \frac{1}{a} \int_0^{+\infty} \frac{\ln(at)}{1+t^2} dt = \frac{1}{a} \int_0^{+\infty} \frac{\ln a}{1+t^2} dt + \frac{1}{a} \int_0^{+\infty} \frac{\ln t}{1+t^2} dt = \frac{\pi \ln a}{2a} + \frac{1}{a} \int_0^{+\infty} \frac{\ln t}{1+t^2} dt. \quad (8)$$

又注意到

$$\int_0^{+\infty} \frac{\ln t}{1+t^2} dt \stackrel{t=\frac{1}{x}}{=} \int_0^{+\infty} \frac{\ln \frac{1}{x}}{1+\frac{1}{x^2}} \frac{1}{x^2} dx = \int_0^{+\infty} \frac{-\ln x}{1+x^2} dx \Rightarrow \int_0^{+\infty} \frac{\ln t}{1+t^2} dt = 0.$$

于是再结合(8)式可得

$$\int_0^{+\infty} \frac{\ln x}{x^2+a^2} dx = \frac{\pi \ln a}{2a}.$$

2.

$$\int_0^\infty \frac{\ln x}{x^2+x+1} dx \stackrel{x=\frac{1}{t}}{=} \int_0^\infty \frac{-\ln t}{1+\frac{1}{t}+\frac{1}{t^2}} d\frac{1}{t} = \int_0^{+\infty} \frac{-\ln t}{1+t+t^2} dt \Rightarrow \int_0^\infty \frac{\ln x}{x^2+x+1} dx = 0.$$

3.

$$\begin{aligned}\int_0^1 \frac{\ln x}{\sqrt{x-x^2}} dx &\stackrel{x=\sin^2 y}{=} \int_0^{\frac{\pi}{2}} \frac{\ln \sin^2 y}{\sqrt{\sin^2 y(1-\sin^2 y)}} d\sin^2 y \\ &= 4 \int_0^{\frac{\pi}{2}} \ln \sin y dy \stackrel{\text{命题 0.3}}{=} 4 \cdot \left(-\frac{\pi}{2} \ln 2 \right) = -2\pi \ln 2.\end{aligned}$$

□

例题 0.4

1. 对 $n \in \mathbb{N}$, 计算 $\int_{-\pi}^\pi \frac{\sin(nx)}{(1+2^x)\sin x} dx.$

2. $\int_{-\pi}^{\pi} \frac{x \sin x \arctan e^x}{1 + \cos^2 x} dx.$
3. 对 $n \in \mathbb{N}$, 计算 $\int_0^{2\pi} \sin(\sin x + nx) dx.$

解 Show/Hide Solution

1.

$$\begin{aligned} \int_{-\pi}^{\pi} \frac{\sin(nx)}{(1+2^x)\sin x} dx &= \int_{-\pi}^0 \left[\frac{\sin(nx)}{(1+2^x)\sin x} + \frac{\sin(nx)}{(1+2^{-x})\sin x} \right] dx = \int_{-\pi}^0 \frac{\sin(nx)}{\sin x} \left(\frac{1}{1+2^x} + \frac{1}{1+2^{-x}} \right) dx \\ &= \int_{-\pi}^0 \frac{\sin(nx)}{\sin x} \cdot \frac{2+2^x+2^{-x}}{2+2^x+2^{-x}} dx = \int_{-\pi}^0 \frac{\sin(nx)}{\sin x} dx = \int_0^{\pi} \frac{\sin(nx)}{\sin x} dx \stackrel{\text{例题 0.1}}{=} \begin{cases} 0, n \text{ 为偶数} \\ \pi, n \text{ 为奇数} \end{cases}. \end{aligned}$$

2.

$$\begin{aligned} \int_{-\pi}^{\pi} \frac{x \sin x \arctan e^x}{1 + \cos^2 x} dx &= \int_{-\pi}^0 \left(\frac{x \sin x \arctan e^x}{1 + \cos^2 x} + \frac{x \sin x \arctan e^{-x}}{1 + \cos^2 x} \right) dx = \int_{-\pi}^0 \frac{x \sin x}{1 + \cos^2 x} (\arctan e^x + \arctan e^{-x}) dx \\ &\stackrel{\text{命题??(1)}}{=} \int_{-\pi}^0 \frac{x \sin x}{1 + \cos^2 x} \cdot \frac{\pi}{2} dx = \frac{\pi}{2} \int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx \\ &= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \left(\frac{x \sin x}{1 + \cos^2 x} + \frac{(\pi-x) \sin x}{1 + \cos^2 x} \right) dx = \frac{\pi^2}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \cos^2 x} dx \\ &= \frac{\pi^2}{2} \arctan \cos x \Big|_0^{\frac{\pi}{2}} = \frac{\pi^2}{2} \cdot \frac{\pi}{4} = \frac{\pi^3}{8}. \end{aligned}$$

3.

$$\begin{aligned} \int_0^{2\pi} \sin(\sin x + nx) dx &= \int_0^{2\pi} \sin[\sin(2\pi - x) + n(2\pi - x)] dx \\ &= \int_0^{2\pi} \sin(-\sin x - nx) dx = - \int_0^{2\pi} \sin(\sin x + nx) dx \\ &\Rightarrow \int_0^{2\pi} \sin(\sin x + nx) dx = 0. \end{aligned}$$

□

例题 0.5 计算 $\int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^{2019})}.$

解 Show/Hide Solution

$$\begin{aligned} \int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^{2019})} &= \int_0^{+\infty} \frac{dx}{(1+x^2)(1+x^{2019})} \\ &\stackrel{t=\frac{1}{x}}{=} \int_0^{+\infty} \frac{t^{2019} dt}{(1+t^2)(1+t^{2019})} = \frac{1}{2} \int_0^{+\infty} \frac{(1+x^{2019}) dx}{(1+x^2)(1+x^{2019})} \\ &= \frac{1}{2} \int_0^{+\infty} \frac{dx}{1+x^2} = \frac{\pi}{4}. \end{aligned}$$

□

0.1.3 Frullani(傅汝兰尼) 积分

定理 0.3 (Frullani(傅汝兰尼) 积分)

设 $f \in C(0, +\infty)$.

1. 若存在极限

$$\lim_{x \rightarrow 0^+} f(x), \quad \lim_{x \rightarrow +\infty} f(x), \quad (9)$$

则对 $a, b > 0$ 有

$$\int_0^{\infty} \frac{f(ax) - f(bx)}{x} dx = \left[\lim_{x \rightarrow 0^+} f(x) - \lim_{x \rightarrow +\infty} f(x) \right] \ln \frac{b}{a}.$$

2. 若存在极限和积分

$$\lim_{x \rightarrow 0^+} f(x) = \alpha, \int_A^\infty \frac{f(x)}{x} dx. \quad (10)$$

则对 $a, b > 0$, 有

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \alpha \ln \frac{b}{a}.$$

3. 若存在极限和积分

$$\lim_{x \rightarrow +\infty} f(x) = \alpha, \int_0^1 \frac{f(x)}{x} dx. \quad (11)$$

则对 $a, b > 0$, 有

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \alpha \ln \frac{a}{b}.$$

4. 若 f 是周期 $T > 0$ 函数且 $\lim_{x \rightarrow 0^+} f(x)$ 存在, 则对 $a, b > 0$ 有

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \left[\lim_{x \rightarrow 0^+} f(x) - \frac{1}{T} \int_0^T f(x) dx \right] \ln \frac{b}{a}.$$

5. 若 f 满足 $\lim_{x \rightarrow 0^+} f(x)$, $\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^x f(y) dy$ 存在, 则对 $a, b > 0$ 有

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \left[\lim_{x \rightarrow 0^+} f(x) - \lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^x f(y) dy \right] \ln \frac{b}{a}.$$



 笔记 Show/Hide Note

傅汝兰尼积分有诸多变种, 无需记忆具体表达式, 知道有大概这么一个东西即可.

证明 Show/Hide Proof

不妨设 $b > a$.

1. 给定 $A > \delta > 0$, 考虑

$$\begin{aligned} \int_\delta^A \frac{f(ax) - f(bx)}{x} dx &= \int_\delta^A \frac{f(ax)}{x} dx - \int_\delta^A \frac{f(bx)}{x} dx \\ &= \int_{a\delta}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{bA} \frac{f(x)}{x} dx \\ &= \int_{bA}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{a\delta} \frac{f(x)}{x} dx \\ &\stackrel{\text{积分中值定理}}{=} f(\theta_1) \int_{bA}^{aA} \frac{1}{x} dx - f(\theta_2) \int_{b\delta}^{a\delta} \frac{1}{x} dx, \end{aligned}$$

这里 $\theta_1 \in (aA, bA), \theta_2 \in (a\delta, b\delta)$, 于是让 $A \rightarrow +\infty, \delta \rightarrow 0^+$, 由(9), 我们知

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \left[\lim_{x \rightarrow 0^+} f(x) - \lim_{x \rightarrow +\infty} f(x) \right] \ln \frac{b}{a}.$$

2. 给定 $A > \delta > 0$, 考虑

$$\begin{aligned} \int_\delta^A \frac{f(ax) - f(bx)}{x} dx &= \int_\delta^A \frac{f(ax)}{x} dx - \int_\delta^A \frac{f(bx)}{x} dx \\ &= \int_{a\delta}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{bA} \frac{f(x)}{x} dx \\ &= \int_{bA}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{a\delta} \frac{f(x)}{x} dx \\ &\stackrel{\text{积分中值定理}}{=} \int_{bA}^{aA} \frac{f(x)}{x} dx - f(\theta) \int_{b\delta}^{a\delta} \frac{1}{x} dx, \end{aligned}$$

这里 $\theta \in (a\delta, b\delta)$, 于是让 $A \rightarrow +\infty, \delta \rightarrow 0^+$, 由 (10), 我们知

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \alpha \ln \frac{b}{a}.$$

3. 给定 $A > \delta > 0$, 考虑

$$\begin{aligned} \int_\delta^A \frac{f(ax) - f(bx)}{x} dx &= \int_\delta^A \frac{f(ax)}{x} dx - \int_\delta^A \frac{f(bx)}{x} dx \\ &= \int_{a\delta}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{bA} \frac{f(x)}{x} dx \\ &= \int_{bA}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{a\delta} \frac{f(x)}{x} dx \\ &\stackrel{\text{积分中值定理}}{=} f(\theta) \int_{bA}^{aA} \frac{1}{x} dx - \int_{b\delta}^{a\delta} \frac{f(x)}{x} dx, \end{aligned}$$

这里 $\theta \in (aA, bA)$, 于是让 $A \rightarrow +\infty, \delta \rightarrow 0^+$, 由 (11), 我们知

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \alpha \ln \frac{a}{b}.$$

4. 给定 $A > \delta > 0$, 考虑

$$\begin{aligned} \int_\delta^A \frac{f(ax) - f(bx)}{x} dx &= \int_\delta^A \frac{f(ax)}{x} dx - \int_\delta^A \frac{f(bx)}{x} dx \\ &= \int_{a\delta}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{bA} \frac{f(x)}{x} dx \\ &= \int_{bA}^{aA} \frac{f(x)}{x} dx - \int_{b\delta}^{a\delta} \frac{f(x)}{x} dx \\ &\stackrel{\text{积分中值定理}}{=} \int_{bA}^{aA} \frac{f(x)}{x} dx - f(\theta) \int_{b\delta}^{a\delta} \frac{1}{x} dx \\ &= \int_b^a \frac{f(Ax)}{x} dx - f(\theta) \int_{b\delta}^{a\delta} \frac{1}{x} dx, \end{aligned}$$

这里 $\theta \in (a\delta, b\delta)$. 现在

$$\lim_{\delta \rightarrow 0^+} \left(-f(\theta) \int_{b\delta}^{a\delta} \frac{1}{x} dx \right) = \lim_{x \rightarrow 0^+} f(x) \ln \frac{b}{a}.$$

由 Riemann 引理, 我们有

$$\lim_{A \rightarrow +\infty} \int_b^a \frac{f(Ax)}{x} dx = \int_b^a \frac{1}{x} dx \cdot \frac{1}{T} \int_0^T f(x) dx = -\frac{1}{T} \int_0^T f(x) dx \cdot \ln \frac{b}{a},$$

这就证明了

$$\int_0^\infty \frac{f(ax) - f(bx)}{x} dx = \left[\lim_{x \rightarrow 0^+} f(x) - \frac{1}{T} \int_0^T f(x) dx \right] \ln \frac{b}{a}.$$

5. 上一问证明中把使用的 Riemann 引理用平均值极限版本的 Riemann 引理代替即可.

□

0.1.4 化成多元累次积分 (换序)

命题 0.4 (重要定积分结论)

证明:

(1) **Gauss - Poisson (高斯-泊松) 积分:** $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$

(2) $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$

(3) $\int_0^{+\infty} \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}.$

(4) **Fresnel(菲涅尔) 积分**: $\int_0^{\infty} \sin x^2 dx = \int_0^{\infty} \cos x^2 dx = \sqrt{\frac{\pi}{8}}.$

证明 Show/Hide Proof

(1) 注意到

$$\begin{aligned} \left(\int_0^{+\infty} e^{-x^2} dx \right)^2 &= \left(\int_0^{+\infty} e^{-x^2} dx \right) \left(\int_0^{+\infty} e^{-y^2} dy \right) = \int_0^{+\infty} e^{-x^2} \left(\int_0^{+\infty} e^{-y^2} dx \right) dy \\ &= \int_0^{+\infty} \left(\int_0^{+\infty} e^{-(x^2+y^2)} dx \right) dy \stackrel{e^{-(x^2+y^2)} \text{连续}}{=} \iint_{R^2} e^{-(x^2+y^2)} dx dy \\ &= \int_0^{\frac{\pi}{2}} d\theta \int_0^{+\infty} r e^{-r^2} dr = \frac{\pi}{2} \int_0^{+\infty} r e^{-r^2} dr \\ &= \frac{\pi}{4} \int_0^{+\infty} e^{-r^2} dr^2 = \frac{\pi}{4}. \end{aligned}$$

$$\text{故 } \int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.$$

(2) 由分部积分可得

$$\begin{aligned} \int_0^{+\infty} e^{-yx} \sin x dx &= e^{-yx} \cos x \Big|_{+\infty}^0 - y \int_0^{+\infty} e^{-yx} \cos x dx = 1 - y \int_0^{+\infty} e^{-yx} \cos x dx \\ &= 1 - y e^{-yx} \sin x \Big|_{+\infty}^0 - y^2 \int_0^{+\infty} e^{-yx} \sin x dx = 1 - y^2 \int_0^{+\infty} e^{-yx} \sin x dx. \end{aligned}$$

因此

$$\int_0^{+\infty} e^{-yx} \sin x dx = \frac{1}{1+y^2}.$$

于是就有

$$\int_0^{+\infty} \frac{\sin x}{x} dx = \int_0^{+\infty} \sin x \left(\int_0^{+\infty} e^{-yx} dy \right) dx = \int_0^{+\infty} dy \int_0^{+\infty} e^{-yx} \sin x dx = \int_0^{+\infty} \frac{1}{y^2+1} dy = \frac{\pi}{2}.$$

(3) 由分部积分可得

$$\begin{aligned} \int_0^{+\infty} \frac{\sin^2 x}{x^2} dx &= \frac{1}{2} \int_0^{+\infty} \frac{1 - \cos 2x}{x^2} dx = -\frac{1}{2} \int_0^{+\infty} (1 - \cos 2x) d\frac{1}{x} \\ &= \frac{1}{2} \frac{1 - \cos x}{x^2} \Big|_{+\infty}^0 + \frac{1}{2} \int_0^{+\infty} \frac{2 \sin 2x}{x} dx \\ &= \int_0^{+\infty} \frac{\sin x}{x} dx \stackrel{\text{命题 0.4(2)}}{=} \frac{\pi}{2}. \end{aligned}$$

(4) 见 Fresnel 积分.

□

命题 0.5

设 $n, m \in \mathbb{N}$, $n \geq m$, 且 $n - m$ 为偶数, 计算 $\int_0^{+\infty} \frac{(\sin x)^n}{x^m} dx$.

解 Show/Hide Solution

(1) 当 $m = n = 1$ 时, 由 **命题 0.4** 知 $\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$

(2) 当 $m = 1, n = 2k - 1$ 时, 由 **定理??** 和 **命题??** 知

$$\begin{aligned} \int_0^{+\infty} \frac{(\sin x)^{2k-1}}{x} dx &\stackrel{\text{定理??}}{=} \frac{1}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin((2k-2i-1)x)}{x} dx \\ &\stackrel{y=(2k-2i-1)x}{=} \frac{1}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin y}{y} dy \end{aligned}$$

$$\begin{aligned}
&= \frac{\pi}{2^{2k-1}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} = (-1)^{k+1} \frac{\pi}{2^{2k-1}} \sum_{i=0}^{k-1} (-1)^i \binom{2k-1}{i} \\
&\stackrel{\text{命题??}}{=} \frac{\pi}{2^{2k-1}} \binom{2k-2}{k-1} = \frac{\pi}{2} \frac{(2k-2)!}{2^{k-1}(k-1)! \cdot 2^{k-1}(k-1)!} \\
&= \frac{\pi}{2} \frac{(2k-2)!!(2k-3)!!}{(2k-2)!! \cdot (2k-2)!!} = \frac{(2k-3)!!}{(2k-2)!!} \cdot \frac{\pi}{2}.
\end{aligned}$$

(3) 当 $m=2, n=2k$ 时, 利用分部积分得

$$\begin{aligned}
\int_0^{+\infty} \frac{(\sin x)^{2k}}{x^2} dx &= -\frac{(\sin x)^{2k}}{x} \Big|_0^{+\infty} + 2k \int_0^{+\infty} \frac{(\sin x)^{2k-1} \cos x}{x} dx = 2k \int_0^{+\infty} \frac{(\sin x)^{2k-1} \cos x}{x} dx \\
&\stackrel{\text{定理??}}{=} \frac{k}{2^{2k-3}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin((2k-2i-1)x) \cos x}{x} dx \\
&= \frac{k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{[\sin((2k-2i)x) + \sin((2k-2i-2)x)]}{x} dx \\
&= \frac{k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin((2k-2i)x)}{x} dx + \frac{k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^{k+i-1} \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin((2k-2i-2)x)}{x} dx \\
&= \frac{(-1)^{k+1}k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^i \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin x}{x} dx + \frac{(-1)^{k+1}k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^i \binom{2k-1}{i} \int_0^{+\infty} \frac{\sin x}{x} dx \\
&= \frac{\pi}{2} \cdot \frac{(-1)^{k+1}k}{2^{2k-2}} \sum_{i=0}^{k-1} (-1)^i \binom{2k-1}{i} \stackrel{\text{命题??}}{=} \frac{\pi}{2} \cdot \frac{k}{2^{2k-2}} \binom{2k-2}{k-1} \\
&= \frac{\pi}{2} \cdot \frac{(2k-2)!}{2^{k-1}(k-1)! \cdot 2^{k-1}(k-1)!} = \frac{\pi}{2} \cdot \frac{(2k-2)!!(2k-3)!!}{(2k-2)!! \cdot (2k-2)!!} \\
&= \frac{(2k-3)!!}{(2k-2)!!} \cdot \frac{\pi}{2}.
\end{aligned}$$

(4) 当 $m > 2$ 时, 利用分部积分得

$$\begin{aligned}
\int_0^{+\infty} \frac{(\sin x)^n}{x^m} dx &= -\frac{1}{m-1} \left[\frac{(\sin x)^n}{x^{m-1}} \Big|_0^{+\infty} - n \int_0^{+\infty} \frac{(\sin x)^{n-1} \cos x}{x^{m-1}} dx \right] \\
&= \frac{n}{m-1} \int_0^{+\infty} \frac{(\sin x)^{n-1} \cos x}{x^{m-1}} dx = \frac{n}{(m-1)(m-2)} \int_0^{+\infty} \frac{[(\sin x)^{n-1} \cos x]'}{x^{m-1}} dx \\
&= \frac{n(n-1)}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^{n-2} \cos^2 x}{x^{m-2}} dx - \frac{n}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^n}{x^{m-2}} dx \\
&= \frac{n(n-1)}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^{n-2} (1 - \sin^2 x)}{x^{m-2}} dx - \frac{n}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^n}{x^{m-2}} dx \\
&= \frac{n(n-1)}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^{n-2}}{x^{m-2}} dx - \frac{n^2}{(m-1)(m-2)} \int_0^{+\infty} \frac{(\sin x)^n}{x^{m-2}} dx.
\end{aligned}$$

不断分部积分, 最后只需计算 $m=1, n=2k-1$ 和 $m=2, n=2k$ ($k \in \mathbb{Z}_+$) 时的情况. 再利用 (2)(3) 的结论即得. □

例题 0.6 计算 $\int_0^1 \sin \ln \frac{1}{x} \cdot \frac{x^b - x^a}{\ln x} dx$ ($b > a > 0$).

证明 [Show/Hide Proof](#)

$$\begin{aligned}
\int_0^1 \sin \ln \frac{1}{x} \cdot \frac{x^b - x^a}{\ln x} dx &= \int_0^1 \sin \ln \frac{1}{x} \left(\int_a^b x^y dy \right) dx = \int_a^b dy \int_0^1 x^y \sin \ln \frac{1}{x} dx \\
&\stackrel{x=e^{-t}}{=} \int_a^b dy \int_{+\infty}^0 e^{-ty} \sin t de^{-t} = \int_a^b dy \int_0^{+\infty} e^{-t(y+1)} \sin t dt
\end{aligned}$$

$$\underline{\underline{\text{命题 0.4(2) 的证明过程}}} \int_a^b \frac{1}{1+(y+1)^2} dy = \arctan(b+1) - \arctan(a+1).$$

□

0.1.5 化成含参积分 (费曼积分法)

例题 0.7 设 $a, b \geq 0$ 且不全为 0, 计算 $\int_0^{\frac{\pi}{2}} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx$.

注 实际上, 根据 $a > b$ 时得到的结果, 可以看出 $F(a, b) = \pi \ln \frac{a+b}{2}$ 对 a, b 有轮换对称性, 故这个结果对其他情况显然也成立.

证明 Show/Hide Proof

设 $F(a, b) = \int_0^{\frac{\pi}{2}} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx$, 当 $a > b$ 时, 则

$$\begin{aligned} \frac{\partial}{\partial b} F(a, b) &= \int_0^{\frac{\pi}{2}} \frac{\partial}{\partial b} \ln(a^2 \cos^2 x + b^2 \sin^2 x) dx = \int_0^{\frac{\pi}{2}} \frac{2b \sin^2 x}{a^2 \cos^2 x + b^2 \sin^2 x} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{2b \tan^2 x}{a^2 + b^2 \tan^2 x} dx = \int_0^{+\infty} \frac{2bt^2}{(a^2 + b^2 t^2)(1+t^2)} dt \\ &= \frac{1}{a^2 - b^2} \int_0^{+\infty} \left(\frac{2a^2 b}{a^2 + b^2 t^2} - \frac{2b}{1+t^2} \right) dt \\ &= \frac{1}{a^2 - b^2} \int_0^{+\infty} \frac{2a^2 b}{a^2 + b^2 t^2} dt - \frac{2b}{a^2 - b^2} \int_0^{+\infty} \frac{1}{1+t^2} dt \\ &= \frac{2b}{a^2 - b^2} \int_0^{+\infty} \frac{1}{1 + (\frac{b}{a}t)^2} dt - \frac{b\pi}{a^2 - b^2} \\ &= \frac{2b}{a^2 - b^2} \cdot \frac{a}{b} \cdot \frac{\pi}{2} - \frac{b\pi}{a^2 - b^2} = \frac{\pi}{a+b}. \end{aligned}$$

于是

$$\begin{aligned} F(a, b) &= F(a, 0) + \int_0^b \frac{\partial}{\partial b'} F(a, b') db' = F(a, 0) + \int_0^b \frac{\pi}{a+b'} db' \\ &= 2 \int_0^{\frac{\pi}{2}} \ln(a \cos x) dx + \pi \ln \frac{a+b}{a} \stackrel{\text{例题 0.3}}{=} \pi \ln \frac{a+b}{2}. \end{aligned}$$

当 $a < b$ 时, 类似可得 $F(a, b) = \pi \ln \frac{a+b}{2}$. 当 $a = b$ 时, $F(a, b) = \int_0^{\frac{\pi}{2}} \ln a^2 dx = \pi \ln a = \pi \ln \frac{a+b}{2}$.

综上, 对 $\forall a, b \geq 0$, 都有 $F(a, b) = \pi \ln \frac{a+b}{2}$.

□

例题 0.8 设 $a > 0, b > 0$, 计算积分 $\int_0^{+\infty} \frac{\ln(a^2 + x^2)}{b^2 + x^2} dx$.

证明 Show/Hide Proof

记 $f(t) = \int_0^{+\infty} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx$, 则

$$\frac{\partial}{\partial t} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} = \frac{x^2}{(b^2 + x^2)(a^2 + tx^2)}.$$

对 $\forall t \in [0, 1]$, 显然 $\frac{\ln(a^2 + tx^2)}{b^2 + x^2}, \frac{x^2}{(b^2 + x^2)(a^2 + tx^2)} \in C[0, 1]$, 且

$$\int_0^1 \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx \leq \int_0^1 \frac{\ln(a^2 + t)}{b^2} dx < +\infty,$$

$$\int_1^{+\infty} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx = \int_1^{+\infty} \frac{\ln\left(\frac{a^2}{x^2} + t\right) + 2 \ln x}{b^2 + x^2} dx \leq \ln(a^2 + t) \int_1^{+\infty} \frac{1}{b^2 + x^2} dx + \int_1^{+\infty} \frac{2 \ln x}{b^2 + x^2} dx < +\infty.$$

故 $f(t) = \int_0^{+\infty} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx$ 在 $(0, 1)$ 上收敛. 注意到对 $\forall [c, d] \subset (0, 1)$, 有

$$\begin{aligned} \int_0^1 \frac{\partial}{\partial t} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx &= \int_0^{+\infty} \frac{x^2}{(b^2 + x^2)(a^2 + tx^2)} dx \\ &= \frac{1}{a^2 - tb^2} \int_0^{+\infty} \left(\frac{a^2}{a^2 + tx^2} - \frac{b^2}{b^2 + x^2} \right) dx \\ &= \frac{1}{a^2 - tb^2} \left[\frac{a}{\sqrt{t}} \arctan \left(\frac{x\sqrt{t}}{a} \right) \Big|_0^{+\infty} - b \arctan \frac{x}{b} \Big|_0^{+\infty} \right] \\ &= \frac{\pi}{2\sqrt{t}(a + b\sqrt{t})} \leq \frac{\pi}{2\sqrt{c}(a + b\sqrt{c})}. \end{aligned}$$

故 $\int_0^1 \frac{\partial}{\partial t} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx$ 在 $(0, 1)$ 上内闭一致收敛. 于是由含参量积分的可微性知

$$\begin{aligned} f'(t) &= \frac{d}{dt} \int_0^{+\infty} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx = \int_0^{+\infty} \frac{\partial}{\partial t} \frac{\ln(a^2 + tx^2)}{b^2 + x^2} dx \\ &= \int_0^{+\infty} \frac{x^2}{(b^2 + x^2)(a^2 + tx^2)} dx = \frac{1}{a^2 - tb^2} \int_0^{+\infty} \left(\frac{a^2}{a^2 + tx^2} - \frac{b^2}{b^2 + x^2} \right) dx \\ &= \frac{1}{a^2 - tb^2} \left[\frac{a}{\sqrt{t}} \arctan \left(\frac{x\sqrt{t}}{a} \right) \Big|_0^{+\infty} - b \arctan \frac{x}{b} \Big|_0^{+\infty} \right] \\ &= \frac{\pi}{2\sqrt{t}(a + b\sqrt{t})}. \end{aligned}$$

故

$$\begin{aligned} \int_0^{+\infty} \frac{\ln(a^2 + x^2)}{b^2 + x^2} dx &= f(1) = \int_0^1 f'(t) dt + f(0) \\ &= \int_0^1 \frac{\pi}{2\sqrt{t}(a + b\sqrt{t})} dt + \int_0^{+\infty} \frac{2 \ln a}{b^2 + x^2} dx \\ &\stackrel{x=\sqrt{t}}{=} \pi \int_0^1 \frac{1}{a + bx} dx + \frac{2 \ln a}{b} \cdot \arctan \frac{x}{b} \Big|_0^{+\infty} \\ &= \frac{\pi}{b} \ln \frac{a+b}{a} + \frac{\pi \ln a}{b} = \frac{\pi}{b} \ln(a+b). \end{aligned}$$

□

0.1.6 级数展开方法

积分和求和换序 $\sum_{n=1}^{\infty} \int_a^b f_n(x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(x) dx$, 等价于

$$\lim_{m \rightarrow \infty} \sum_{n=1}^m \int_a^b f_n(x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(x) dx.$$

又由于有限和随意交换, 因此上式等价于

$$\lim_{m \rightarrow \infty} \int_a^b \sum_{n=1}^m f_n(x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(x) dx,$$

于是

$$\sum_{n=1}^{\infty} \int_a^b f_n(x) dx = \int_a^b \sum_{n=1}^{\infty} f_n(x) dx \iff \lim_{m \rightarrow \infty} \int_a^b \sum_{n=m+1}^{\infty} f_n(x) dx = 0.$$

例题 0.9 计算 $\int_0^{\infty} \frac{x}{1+e^x} dx$.

解 Show/Hide Solution

由于

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad 0 < x < 1.$$

并且 $0 < e^{-x} < 1$, 故

$$\begin{aligned} \int_0^{+\infty} \frac{x}{1+e^x} dx &= \int_0^{+\infty} \frac{x e^{-x}}{1+e^{-x}} dx = \int_0^{+\infty} x \sum_{n=0}^{\infty} (-1)^n e^{-(n+1)x} dx \\ &\stackrel{\text{换序}}{=} \sum_{n=0}^{\infty} (-1)^n \int_0^{+\infty} x e^{-(n+1)x} dx = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n+1)^2} \\ &= \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} - \sum_{n=0}^{\infty} \frac{1}{(2n)^2}. \end{aligned} \quad (12)$$

又因为 $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, 所以

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{1}{(2n)^2} &= \frac{1}{4} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{24}, \\ \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} &= \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=0}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{8}. \end{aligned}$$

故

$$\int_0^{+\infty} \frac{x}{1+e^x} dx = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} - \sum_{n=0}^{\infty} \frac{1}{(2n)^2} = \frac{\pi^2}{8} - \frac{\pi^2}{24} = \frac{\pi^2}{12}.$$

下面证明(12)式换序成立, 等价于证明 $\lim_{m \rightarrow +\infty} \int_0^{+\infty} \sum_{n=m}^{\infty} x(-1)^n e^{-(n+1)x} dx = 0$. 由交错级数不等式及 $x e^{-(n+1)x}$ 关于 n 非负递减, 对 $\forall m \in \mathbb{N}$, 都有

$$\int_0^{+\infty} \left| \sum_{n=m}^{\infty} x(-1)^n e^{-(n+1)x} \right| dx \leq \int_0^{+\infty} x e^{-(m+1)x} dx = -\frac{x e^{-(m+1)x}}{m+1} \Big|_0^{+\infty} + \frac{1}{m+1} \int_0^{+\infty} e^{-(m+1)x} dx = \frac{1}{(m+1)^2}.$$

令 $m \rightarrow +\infty$, 得 $\lim_{m \rightarrow +\infty} \int_0^{+\infty} \sum_{n=m}^{\infty} x(-1)^n e^{-(n+1)x} dx = 0$. 故(12)式换序成立.

□

命题 0.6

证明:

- (1) $\sum_{n=1}^{\infty} \frac{q^n \sin(nx)}{n} = \arctan \frac{q \sin x}{1 - q \cos x}, |q| \leq 1.$
- (2) $\sum_{n=1}^{\infty} \frac{q^n \cos(nx)}{n} = -\frac{1}{2} \ln(1 + q^2 - 2q \cos x), |q| \leq 1.$
- (3) $\sum_{n=1}^{\infty} \frac{q^n \cos(nx)}{n!} = e^{q \cos x} \cos(q \sin x) - 1, |q| \leq 1, x \in \mathbb{R}.$
- (4) $\sum_{n=1}^{\infty} \frac{q^n \sin(nx)}{n!} = e^{q \cos x} \sin(q \sin x), |q| \leq 1, x \in \mathbb{R}.$

笔记 Show/Hide Note

在 $\mathbb{C}$ 上,

$$\operatorname{Ln} z = \ln |z| + i(\arg z + 2k\pi), k \in \mathbb{Z}.$$

我们定义主值支

$$\ln z = \ln |z| + i \arg z.$$

本部分内容无需记忆,只需要大概有个可以算的感觉即可,实际做题中可以围绕这种级数给出构造.

证明 Show/Hide Proof

$\Im$ 表示取虚部, $\Re$ 表示取实部.

(1) 利用欧拉公式有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n \sin(nx)}{n} &= \Im \left(\sum_{n=1}^{\infty} \frac{q^n e^{inx}}{n} \right) = \Im \left(\sum_{n=1}^{\infty} \frac{(qe^{ix})^n}{n} \right) = \Im(-\ln(1 - qe^{ix})) \\ &= -\Im \left(\ln |1 - qe^{ix}| + i \frac{-q \sin x}{1 - q \cos x} \right) = \arctan \frac{q \sin x}{1 - q \cos x}. \end{aligned}$$

(2) 利用欧拉公式有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n \cos(nx)}{n} &= -\Re \left(\ln |1 - qe^{ix}| + i \frac{-q \sin x}{1 - q \cos x} \right) = -\frac{1}{2} \ln [(1 - q \cos x)^2 + q^2 \sin^2 x] \\ &= -\frac{1}{2} \ln(1 + q^2 - 2q \cos x). \end{aligned}$$

(3) 利用欧拉公式有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n \cos(nx)}{n!} &= \Re \left(\sum_{n=1}^{\infty} \frac{(qe^{ix})^n}{n!} \right) = \Re(e^{qe^{ix}} - 1) = \Re(e^{q \cos x + iq \sin x} - 1) \\ &= \Re(e^{q \cos x} \cos(q \sin x) - 1 + ie^{q \cos x} \sin(q \sin x)) \\ &= e^{q \cos x} \cos(q \sin x) - 1. \end{aligned}$$

(4) 利用 (3) 有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{q^n \sin(nx)}{n!} &= \Im(e^{q \cos x} \cos(q \sin x) - 1 + ie^{q \cos x} \sin(q \sin x)) \\ &= e^{q \cos x} \sin(q \sin x). \end{aligned}$$

□

例题 0.10 计算

1. $\int_0^{2\pi} e^{\cos x} \cos(\sin x) dx.$
2. $\int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx, a \in (0, +\infty) \setminus \{1\}.$

注 由 1 的证明可得

$$e^{\cos x} \cos(\sin x) = \operatorname{Re} \left(\sum_{n=0}^{\infty} \frac{(e^{ix})^n}{n!} \right) = \operatorname{Re} \left(\sum_{n=0}^{\infty} \frac{e^{inx}}{n!} \right) = \sum_{n=0}^{\infty} \frac{\cos(nx)}{n!}.$$

实际上, 上式就是命题 0.6(3) 的结论.

注 第 2 问也可以用含参积分求导的方法进行计算 (这个方法更容易想到).

证明 Show/Hide Proof

1.

$$\begin{aligned} \int_0^{2\pi} e^{\cos x} \cos(\sin x) dx &= \operatorname{Re} \left(\int_0^{2\pi} e^{\cos x} e^{i \sin x} dx \right) = \operatorname{Re} \left(\int_0^{2\pi} e^{\cos x + i \sin x} dx \right) \\ &= \operatorname{Re} \left(\int_0^{2\pi} e^{e^{ix}} dx \right) = \operatorname{Re} \left[\int_0^{2\pi} \sum_{n=0}^{+\infty} \frac{(e^{ix})^n}{n!} dx \right] = \operatorname{Re} \left[\sum_{n=0}^{+\infty} \int_0^{2\pi} \frac{(e^{ix})^n}{n!} dx \right] \\ &= \operatorname{Re} \left(\sum_{n=0}^{+\infty} \int_0^{2\pi} \frac{e^{inx}}{n!} dx \right) = \operatorname{Re} \left(\int_0^{2\pi} \frac{e^{i \cdot 0 \cdot x}}{0!} dx + \sum_{n=1}^{+\infty} \frac{e^{2\pi i x} - 1}{in \cdot n!} \right) \end{aligned}$$

$$= \operatorname{Re} \left(\int_0^{2\pi} 1 dx + 0 \right) = 2\pi.$$

2. 注意到当 $a \in (0, 1)$ 时, 有

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{a^n \cos(nx)}{n} &= \operatorname{Re} \left[\sum_{n=1}^{\infty} \frac{(ae^{ix})^n}{n} \right] = -\operatorname{Re} [\ln(1 - ae^{ix})] \\ &= -\operatorname{Re} [\ln|1 - ae^{ix}| + i \arg(1 - ae^{ix})] = -\ln|1 - ae^{ix}| \\ &= -\ln|(1 - a \cos x) + ai \sin x| = -\frac{1}{2} \ln(1 + a^2 - 2a \cos x). \end{aligned}$$

于是当 $a \in (0, 1)$ 时, 就有

$$\int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx = -\frac{1}{2} \int_0^{\pi} \sum_{n=1}^{\infty} \frac{a^n \cos(nx)}{n} dx = 0.$$

若 $a > 1$, 则 $\frac{1}{a} \in (0, 1)$, 从而此时我们有

$$\int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx = \pi \ln a^2 + \int_0^{\pi} \ln \left(\frac{1}{a^2} - \frac{2}{a} \cos x + 1 \right) dx = \pi \ln a^2 = 2\pi \ln a.$$

又由 $\ln(1 - 2a \cos x + a^2)$ 关于 a 的偏导存在可知 $\int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx$ 关于 a 连续. 于是由

$$\int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx = 2\pi \ln a, \quad \forall a > 1.$$

可知当 $a = 1$ 时, 我们有

$$\int_0^{\pi} \ln(2 - 2 \cos x) dx = \lim_{a \rightarrow 1^+} \int_0^{\pi} \ln(1 - 2a \cos x + a^2) dx = \lim_{a \rightarrow 1^+} (2\pi \ln a) = 0.$$

□

定义 0.1 (多重对数函数-Li₂ 函数)

定义

$$\operatorname{Li}_2(x) \triangleq \sum_{n=1}^{\infty} \frac{x^n}{n^2}, \quad x \in [-1, 1].$$

♣

命题 0.7 (Li₂ 函数的基本性质)

$$(1) \operatorname{Li}_2(x) + \operatorname{Li}_2(1-x) = \frac{\pi^2}{6} - \ln x \cdot \ln(1-x), x \in (0, 1).$$

$$(2) \operatorname{Li}_2(1) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}, \quad \operatorname{Li}_2(0) = 0, \quad \operatorname{Li}_2\left(\frac{1}{2}\right) = \frac{\pi^2}{12} - \frac{1}{2} \ln^2 \frac{1}{2}.$$

♠

证明 Show/Hide Proof

(1) 记 $f(x) \triangleq \operatorname{Li}_2(x)$, $F(x) \triangleq f(x) + f(1-x) + \ln x \ln(1-x)$. 显然 $f(x)$ 收敛域为 $[-1, 1]$, 故由函数项级数的逐项可微性知

$$f'(x) = \sum_{n=1}^{\infty} \frac{x^{n-1}}{n} = -\frac{1}{x} \ln(1-x).$$

于是

$$F'(x) = -\frac{1}{x} \ln(1-x) + \frac{\ln x}{1-x} - \frac{\ln x}{1-x} + \frac{\ln(1-x)}{x} = 0.$$

$$\text{故 } F(x) \equiv F(1) = f(0) + f(1) = 0 + \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

(2) 显然 $\text{Li}_2(1) = \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$, $\text{Li}_2(0) = 0$. 由 (1) 可得

$$\text{Li}_2\left(\frac{1}{2}\right) + \text{Li}_2\left(\frac{1}{2}\right) = 2\text{Li}_2\left(\frac{1}{2}\right) = \frac{\pi^2}{6} - \ln^2 \frac{1}{2} \implies \text{Li}_2\left(\frac{1}{2}\right) = \frac{\pi^2}{12} - \frac{1}{2} \ln^2 \frac{1}{2}.$$

□

例题 0.11 计算 $\int_0^{\frac{1}{2}} \frac{\ln x}{1-x} dx$.

解 Show/Hide Solution

$$\begin{aligned} \int_0^{\frac{1}{2}} \frac{\ln x}{1-x} dx &= \int_{\frac{1}{2}}^1 \frac{\ln(1-x)}{x} dx = - \sum_{n=1}^{\infty} \frac{1}{n} \int_{\frac{1}{2}}^1 x^{n-1} dx \\ &= - \sum_{n=1}^{\infty} \frac{1}{n^2} + \sum_{n=1}^{\infty} \frac{1}{2^n n^2} = -\frac{\pi^2}{6} + \text{Li}_2\left(\frac{1}{2}\right) \\ &\quad \underline{\text{Li}_2 \text{ 函数的基本性质}} -\frac{\pi^2}{12} - \frac{1}{2} \ln^2 \frac{1}{2}. \end{aligned}$$

□

0.1.7 重积分计算

定理 0.4 (二重积分换序)

证明:

$$\int_a^b dx \int_a^x f(x, y) dy = \int_a^b dy \int_y^b f(x, y) dx, \quad (10)$$

其中 $f(x, y)$ 是在由直线 $y = a, x = b, y = x$ 所围成的三角形 (Δ) 上连续的任意函数.

♥

证明 Show/Hide Proof

□

命题 0.8

设 $f(x)$ 在 $[a, b]$ 上连续, 试证: 对任意 $x \in (a, b)$, 有

$$\int_a^x dx_1 \int_a^{x_1} dx_2 \cdots \int_a^{x_n} f(x_{n+1}) dx_{n+1} = \frac{1}{n!} \int_a^x (x-y)^n f(y) dy, \quad n = 1, 2, \dots$$

♣

证明 Show/Hide Proof

当 $n = 1$ 时, 由二重积分换序可知

$$\int_a^x dx_1 \int_a^{x_1} f(x_2) dx_2 = \int_a^x dx_2 \int_{x_2}^x f(x_2) dx_1 = \int_a^x (x - x_2) f(x_2) dx_2 = \int_a^x (x - y) f(y) dy.$$

设原结论对 $n = k - 1$ 的情形成立, 考虑 $n = k$ 的情形, 由归纳假设可知

$$\int_a^{x_1} dx_1 \cdots \int_a^{x_k} f(x_{k+1}) dx_{k+1} = \frac{1}{(k-1)!} \int_a^{x_1} (x_1 - y)^{k-1} f(y) dy.$$

于是再利用二重积分换序得

$$\begin{aligned} \int_a^x dx_1 \int_a^{x_1} dx_2 \cdots \int_a^{x_k} f(x_{k+1}) dx_{k+1} &= \frac{1}{(k-1)!} \int_a^x dx_1 \int_a^{x_1} (x_1 - y)^{k-1} f(y) dy \\ &= \frac{1}{(k-1)!} \int_a^x dy \int_y^x (x_1 - y)^{k-1} f(y) dx_1 \\ &= \frac{1}{k!} \int_a^x (x - y)^k f(y) dy. \end{aligned}$$

故由数学归纳法知原结论成立.

例题 0.12 求定义在星形区域 $D = \{(x, y) \mid x^{\frac{2}{3}} + y^{\frac{2}{3}} \leq 1\}$ 上满足 $f(1, 0) = 1$ 的正值连续函数 f 使得 $\iint_D \frac{f(x, y)}{f(y, x)} dx dy$ 达到最小, 并求出这个最小值. □

解 Show/Hide Solution

对积分 $I = \iint_D \frac{f(x, y)}{f(y, x)} dx dy$ 作变换 $x \rightarrow y, y \rightarrow x$, 由 D 的对称性, 知 $I = \iint_D \frac{f(y, x)}{f(x, y)} dx dy$. 因而由均值不等式可得

$$I = \frac{1}{2} \iint_D \left(\frac{f(x, y)}{f(y, x)} + \frac{f(y, x)}{f(x, y)} \right) dx dy \geq \iint_D 1 dx dy = \sigma(D),$$

这里 $\sigma(D)$ 是 D 的面积.

$$I - \sigma(D) = \frac{1}{2} \iint_D \left(\sqrt{\frac{f(x, y)}{f(y, x)}} - \sqrt{\frac{f(y, x)}{f(x, y)}} \right)^2 dx dy \geq 0.$$

$I = \sigma(D)$ 当且仅当 $f(x, y) = f(y, x)$. 故所求函数为所有满足 $f(x, y) = f(y, x)$ 及 $f(1, 0) = 1$ 的连续正值函数.

D 的边界的参数方程为

$$x = \cos^3 \varphi, \quad y = \sin^3 \varphi \quad (0 \leq \varphi \leq 2\pi),$$

故 I 的最小值为

$$\begin{aligned} \sigma(D) &= \iint_D 1 dx dy = 4 \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq \frac{\pi}{2}}} 3r \sin^2 \varphi \cos^2 \varphi dr d\varphi \\ &= 6 \int_0^{\frac{\pi}{2}} \sin^2 \varphi \cos^2 \varphi d\varphi = \frac{3}{8} \pi. \end{aligned}$$

所以所求最小值是 $\frac{3}{8}\pi$, 且当 $f(x, y) = f(y, x)$ 并满足 $f(1, 0) = 1$ 时, 取到该最小值. □

例题 0.13 求证: $\iint_{[0,1]^2} (xy)^{xy} dx dy = \int_0^1 t^t dt$.

证明 Show/Hide Proof

首先化为累次积分

$$\iint_{[0,1]^2} (xy)^{xy} dx dy = \int_0^1 dx \int_0^1 (xy)^{xy} dy = \int_0^1 dx \int_0^x \frac{t^t}{x} dt = \int_0^1 \frac{f(x)}{x} dx,$$

其中 $f(x) = \int_0^x t^t dt$. 由分部积分,

$$\int_0^1 \frac{f(x)}{x} dx = f(x) \ln x \Big|_0^1 - \int_0^1 x^x \ln x dx = - \int_0^1 x^x \ln x dx.$$

因为 $(x^x)' = x^x \ln x + x^x$, 所以

$$\int_0^1 x^x \ln x dx = \int_0^1 ((x^x)' - x^x) dx = - \int_0^1 x^x dx.$$

于是

$$\iint_{[0,1]^2} (xy)^{xy} dx dy = \int_0^1 t^t dt.$$

例题 0.14 计算二重积分 $I = \iint_D \operatorname{sgn}(x^2 - y^2 + 2) dx dy$, 其中 $D = \{(x, y) \mid x^2 + y^2 \leq 4\}$. □

解 Show/Hide Solution

设 D 在第一象限部分为 D_1 , 则由对称性

$$I = 4 \iint_{D_1} \operatorname{sgn}(x^2 - y^2 + 2) \, dx dy.$$

设 D_2 是 D_1 中使得 $x^2 - y^2 + 2 < 0$ 的部分, D_3 是 D_1 中使得 $x^2 - y^2 + 2 \geq 0$ 的部分, 则 $D_1 = D_2 \cup D_3$. 因此

$$\begin{aligned} I &= 4 \left[\iint_{D_3} dx dy - \iint_{D_2} dx dy \right] = 4[\sigma(D_3) - \sigma(D_2)] \\ &= 4 \left[\frac{1}{4} \cdot \pi \cdot 2^2 - 2\sigma(D_2) \right] = 4\pi - 8\sigma(D_2), \end{aligned}$$

其中 $\sigma(D_2), \sigma(D_3)$ 分别表示 D_2 和 D_3 的面积. 在极坐标 $x = r \cos \varphi, y = r \sin \varphi$ 之下, D_2 为

$$\left\{ (r, \varphi) \mid \frac{\pi}{3} \leq \varphi \leq \frac{\pi}{2}, \sqrt{-\frac{2}{\cos 2\varphi}} \leq r \leq 2 \right\}.$$

因而

$$\begin{aligned} \sigma(D_2) &= \iint_{D_2} dx dy = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} d\varphi \int_{\sqrt{-\frac{2}{\cos 2\varphi}}}^2 r \, dr \\ &= \frac{1}{2} \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \left(4 + \frac{2}{\cos 2\varphi} \right) d\varphi = \frac{\pi}{3} + \frac{1}{2} \int_{\frac{2\pi}{3}}^{\pi} \frac{1}{\cos \varphi} d\varphi \\ &= \frac{\pi}{3} - \frac{1}{2} \ln(2 + \sqrt{3}), \end{aligned}$$

故

$$I = \frac{4\pi}{3} + 4 \ln(2 + \sqrt{3}).$$

□

例题 0.15 设 $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. 求 $I = \iint_D \left| \frac{x+y}{\sqrt{2}} - x^2 - y^2 \right| dx dy$.

解 Show/Hide Solution

由极坐标变换 $x = r \cos \varphi, y = r \sin \varphi, 0 \leq r \leq 1, 0 \leq \varphi \leq 2\pi$, 有

$$\begin{aligned} I &= \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq 2\pi}} \left| \frac{\cos \varphi + \sin \varphi}{\sqrt{2}} - r \right| r^2 dr d\varphi \\ &= \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq 2\pi}} \left| \sin \left(\varphi + \frac{\pi}{4} \right) - r \right| r^2 dr d\varphi = \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq 2\pi}} |\sin \varphi - r| r^2 dr d\varphi \\ &= \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq \pi}} |\sin \varphi - r| r^2 dr d\varphi + \iint_{\substack{0 \leq r \leq 1 \\ \pi \leq \varphi \leq 2\pi}} |\sin \varphi - r| r^2 dr d\varphi \\ &= \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq \pi}} |\sin \varphi - r| r^2 dr d\varphi + \iint_{\substack{0 \leq r \leq 1 \\ 0 \leq \varphi \leq \pi}} (\sin \varphi + r) r^2 dr d\varphi. \end{aligned}$$

因此, 有

$$\begin{aligned} I &= \int_0^\pi d\varphi \int_0^{\sin \varphi} (\sin \varphi - r) r^2 dr + \int_0^\pi d\varphi \int_{\sin \varphi}^1 (r - \sin \varphi) r^2 dr \\ &\quad + \int_0^\pi d\varphi \int_0^{\sin \varphi} (\sin \varphi + r) r^2 dr + \int_0^\pi d\varphi \int_{\sin \varphi}^1 (\sin \varphi + r) r^2 dr \\ &= \int_0^\pi d\varphi \int_0^{\sin \varphi} 2 \sin \varphi \cdot r^2 dr + \int_0^\pi d\varphi \int_{\sin \varphi}^1 2r \cdot r^2 dr \end{aligned}$$

$$\begin{aligned}
&= \int_0^\pi \frac{2}{3} \sin^4 \varphi d\varphi + \int_0^\pi \frac{1}{2} (1 - \sin^4 \varphi) d\varphi \\
&= \frac{1}{6} \int_0^\pi \sin^4 \varphi d\varphi + \frac{\pi}{2} = \frac{1}{6} \cdot \frac{3\pi}{8} + \frac{\pi}{2} = \frac{9}{16}\pi.
\end{aligned}$$

□

例题 0.16 设 f 是定义在正方形 $S = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$ 上的四阶连续可微函数, 在 S 的边界上为零, 并且

$$\left| \frac{\partial^4 f}{\partial x^2 \partial y^2} \right| \leq M.$$

求证:

$$\left| \iint_S f(x, y) dx dy \right| \leq \frac{1}{144} M.$$

证明 Show/Hide Proof

考虑函数 $g(x, y) = x(1-x)y(1-y)$. 易知

$$\frac{\partial^4 g}{\partial x^2 \partial y^2} = 4, \quad \iint_S g(x, y) dx dy = \frac{1}{36}.$$

因为 f 在 S 的边界上为零, 所以 $\frac{\partial^2 f}{\partial y^2}$ 在 $x=0$ 和 $x=1$ 时为零. 于是

$$\begin{aligned}
\iint_S \frac{\partial^4 f}{\partial x^2 \partial y^2} \cdot g dx dy &= \int_0^1 dy \int_0^1 \frac{\partial^4 f}{\partial x^2 \partial y^2} \cdot g dx \\
&= \int_0^1 dy \left(\left. \frac{\partial^3 f}{\partial x \partial y^2} \cdot g \right|_{x=0}^1 - \int_0^1 \frac{\partial^3 f}{\partial x \partial y^2} \cdot \frac{\partial g}{\partial x} dx \right) \\
&= - \int_0^1 dy \int_0^1 \frac{\partial^3 f}{\partial x \partial y^2} \cdot \frac{\partial g}{\partial x} dx \\
&= - \int_0^1 dy \left(\left. \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial g}{\partial x} \right|_{x=0}^1 - \int_0^1 \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial^2 g}{\partial x^2} dx \right) \\
&= \int_0^1 dy \int_0^1 \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial^2 g}{\partial x^2} dx \\
&= \iint_S \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial^2 g}{\partial x^2} dx dy.
\end{aligned}$$

同理, 由于 $\frac{\partial^2 g}{\partial x^2}$ 在 $y=0$ 和 $y=1$ 时为零, 作与上面类似的推导, 可得

$$\iint_S \frac{\partial^4 g}{\partial x^2 \partial y^2} \cdot f dx dy = \iint_S \frac{\partial^2 f}{\partial y^2} \cdot \frac{\partial^2 g}{\partial x^2} dx dy.$$

因此

$$\iint_S \frac{\partial^4 f}{\partial x^2 \partial y^2} \cdot g dx dy = \iint_S \frac{\partial^4 g}{\partial x^2 \partial y^2} \cdot f dx dy.$$

从而

$$\begin{aligned}
\left| \iint_S f dx dy \right| &= \frac{1}{4} \left| \iint_S 4f dx dy \right| = \frac{1}{4} \left| \iint_S \frac{\partial^4 g}{\partial x^2 \partial y^2} f dx dy \right| \\
&= \frac{1}{4} \left| \iint_S \frac{\partial^4 f}{\partial x^2 \partial y^2} \cdot g dx dy \right| \leq \frac{M}{4} \iint_S g dx dy = \frac{M}{144}.
\end{aligned}$$

□

定理 0.5 (Poincaré(庞加莱) 不等式)

设 φ, ψ 是 $[a, b]$ 上的连续函数, f 在区域 $D = \{(x, y) \mid a \leq x \leq b, \varphi(x) \leq y \leq \psi(x)\}$ 上连续可微, 且有 $f(x, \varphi(x)) = 0$ ($x \in [a, b]$). 则存在 $M > 0$, 使得

$$\iint_D f^2(x, y) \, dx dy \leq M \iint_D (f'_y(x, y))^2 \, dx dy.$$

证明 Show/Hide Proof

由 Newton-Leibniz 公式和 Cauchy 不等式可得

$$\begin{aligned} f^2(x, y) &= [f(x, y) - f(x, \varphi(x))]^2 = \left(\int_{\varphi(x)}^y \frac{\partial f}{\partial t}(x, t) \, dt \right)^2 \\ &\leq (y - \varphi(x)) \int_{\varphi(x)}^y \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt, \end{aligned}$$

因此

$$\begin{aligned} \iint_D f^2(x, y) \, dx dy &= \int_a^b dx \int_{\varphi(x)}^{\psi(x)} f^2(x, y) \, dy \\ &\leq \int_a^b dx \int_{\varphi(x)}^{\psi(x)} (y - \varphi(x)) \, dy \int_{\varphi(x)}^y \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt \\ &= \int_a^b dx \int_{\varphi(x)}^{\psi(x)} \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt \int_t^{\psi(x)} (y - \varphi(x)) \, dy \\ &\leq \int_a^b dx \int_{\varphi(x)}^{\psi(x)} \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt \int_{\varphi(x)}^{\psi(x)} (y - \varphi(x)) \, dy \\ &= \int_a^b dx \int_{\varphi(x)}^{\psi(x)} \frac{1}{2} (\psi(x) - \varphi(x))^2 \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt \\ &\leq M \int_a^b \left(\int_{\varphi(x)}^{\psi(x)} \left(\frac{\partial f}{\partial t}(x, t) \right)^2 \, dt \right) dx \\ &= M \iint_D \left(\frac{\partial f}{\partial y}(x, y) \right)^2 \, dx dy, \end{aligned}$$

这里 M 是满足 $M > \max_{a \leq x \leq b} \frac{1}{2} (\psi(x) - \varphi(x))^2$ 的常数.

□

例题 0.17 设 $a > 0$, $\Omega_n(a) : x_1 + x_2 + \cdots + x_n \leq a, x_i \geq 0$ ($i = 1, 2, \dots, n$). 求积分

$$I_n(a) = \int \cdots \int_{\Omega_n(a)} x_1 x_2 \cdots x_n \, dx_1 dx_2 \cdots dx_n.$$

解 Show/Hide Solution

作变换 $x_i = at_i, i = 1, 2, \dots, n$, 则

$$I_n(a) = a^{2n} \int \cdots \int_{\Omega_n(1)} t_1 t_2 \cdots t_n \, dt_1 dt_2 \cdots dt_n = a^{2n} I_n(1).$$

再用累次积分, 可得

$$\begin{aligned} I_n(1) &= \int \cdots \int_{\Omega_n(1)} t_1 t_2 \cdots t_n \, dt_1 dt_2 \cdots dt_n \\ &= \int_0^1 t_n \, dt_n \int \cdots \int_{t_1 + t_2 + \cdots + t_{n-1} \leq 1 - t_n} t_1 \cdots t_{n-1} \, dt_1 \cdots dt_{n-1} \\ &= \int_0^1 t_n I_{n-1}(1 - t_n) \, dt_n = \int_0^1 t_n (1 - t_n)^{2(n-1)} I_{n-1}(1) \, dt_n. \end{aligned}$$

因此,

$$I_n(1) = \frac{1}{2n(2n-1)} I_{n-1}(1).$$

注意到 $I_1(1) = \int_0^1 t dt = \frac{1}{2}$. 由上面的递推公式, 可得 $I_n(1) = \frac{1}{(2n)!}$. 故 $I_n(a) = \frac{a^{2n}}{(2n)!}$.

□

0.1.8 其他

例题 0.18 证明积分 $\int_0^{\infty} e^{-ax^2 - \frac{b}{x^2}} dx = \frac{1}{2} \sqrt{\frac{\pi}{a}} e^{-2\sqrt{ab}}, a, b > 0$.

证明 Show/Hide Proof

当 $a = 1$ 时, 就有

$$\begin{aligned} \int_0^{+\infty} e^{-x^2 - \frac{b}{x^2}} dx &= e^{-2\sqrt{b}} \int_0^{+\infty} e^{-\left(x - \frac{\sqrt{b}}{x}\right)^2} dx \stackrel{y = \frac{\sqrt{b}}{x}}{=} e^{-2\sqrt{b}} \int_0^{+\infty} \frac{\sqrt{b}}{y^2} e^{-\left(\frac{\sqrt{b}}{y} - y\right)^2} dy \\ &= \frac{e^{-2\sqrt{b}}}{2} \int_0^{+\infty} \left(1 + \frac{\sqrt{b}}{y^2}\right) e^{-\left(y - \frac{\sqrt{b}}{y}\right)^2} dy = \frac{e^{-2\sqrt{b}}}{2} \int_0^{+\infty} e^{-\left(y - \frac{\sqrt{b}}{y}\right)^2} d\left(y - \frac{\sqrt{b}}{y}\right) \\ &= \frac{e^{-2\sqrt{b}}}{2} \int_{-\infty}^{+\infty} e^{-t^2} dt = \frac{\sqrt{\pi}}{2} e^{-2\sqrt{b}}. \end{aligned}$$

于是对 $\forall a > 0$, 就有

$$\int_0^{+\infty} e^{-ax^2 - \frac{b}{x^2}} dx = \frac{1}{\sqrt{a}} \int_0^{+\infty} e^{-x^2 - \frac{ab}{x^2}} dx = \frac{1}{\sqrt{a}} \sqrt{\frac{\pi}{a}} e^{-2\sqrt{ab}}.$$

□

例题 0.19 计算 $\int_0^{\infty} \frac{\cos(ax)}{1+x^2} dx, a \in \mathbb{R}$.

注 本题可以用复变函数的方法 (留数定理) 来计算. 但是我们这里用基本的高等数学的方法来计算.

$\int_0^{\infty} \frac{\sin(ax)}{1+x^2} dx$ 这个积分没办法算出具体的初等数值.

证明 Show/Hide Proof

$$\begin{aligned} \int_0^{+\infty} \frac{\cos(ax)}{1+x^2} dx &= \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos(ax)}{1+x^2} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \cos(ax) \left(\int_0^{+\infty} e^{-(1+x^2)y} dy \right) dx \\ &= \frac{1}{2} \int_{-\infty}^{+\infty} \left(\int_0^{+\infty} e^{-(1+x^2)y} \cos(ax) dy \right) dx = \frac{1}{2} \int_{-\infty}^{+\infty} \int_0^{+\infty} e^{-(1+x^2)y} \cos(ax) dy dx \\ &= \frac{1}{2} \int_0^{+\infty} \left(\int_{-\infty}^{+\infty} e^{-(1+x^2)y} \cos(ax) dx \right) dy = \frac{1}{2} \int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-x^2 y} \cos(ax) dx \right) dy \\ &= \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-x^2 y + iax} dx \right) dy \right) = \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-y \left(x^2 - \frac{aix}{y} + \left(\frac{ai}{2y} \right)^2 \right) - \frac{a^2}{4y}} dx \right) dy \right) \\ &= \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-y \left(x - \frac{ai}{2y} \right)^2 - \frac{a^2}{4y}} dx \right) dy \right) = \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-y \left(x + \frac{a}{2iy} \right)^2 - \frac{a^2}{4y}} dx \right) dy \right) \\ &= \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-y \left(x + \frac{a}{2iy} \right)^2 - \frac{a^2}{4y}} d \left(x + \frac{a}{2iy} \right) \right) dy \right) = \frac{1}{2} \operatorname{Re} \left(\int_0^{+\infty} e^{-y} \left(\int_{-\infty}^{+\infty} e^{-yx^2 - \frac{a^2}{4y}} dx \right) dy \right) \\ &= \frac{1}{2} \int_0^{+\infty} e^{-y - \frac{a^2}{4y}} \left(\int_{-\infty}^{+\infty} e^{-yx^2} dx \right) dy = \frac{\sqrt{\pi}}{2} \int_0^{+\infty} \frac{1}{\sqrt{y}} e^{-y - \frac{a^2}{4y}} dy \\ &\stackrel{y=t^2}{=} \sqrt{\pi} \int_0^{+\infty} e^{-t^2 - \frac{a^2}{4t^2}} dt \stackrel{\text{例题 0.18}}{=} \sqrt{\pi} \cdot \frac{\sqrt{\pi}}{2} e^{-|a|} = \frac{\pi}{2} e^{-|a|}. \end{aligned}$$

□

例题 0.20 计算 $\int_0^{\infty} \frac{1}{(1+x^8)^2} dx$.

注 由命题??可知对 $\forall s > 0$, 都有

$$\frac{1}{t^s} = \frac{1}{\Gamma(s)} \int_0^{+\infty} y^{s-1} e^{-ty} dy, \forall t \in \mathbb{R}.$$

本题的核心想法就是利用上式将 $\frac{1}{1+x^8}$ 转化成积分形式.

证明 Show/Hide Proof

注意到

$$\int_0^{+\infty} y e^{-(1+x^8)y} dy \stackrel{y=\frac{z}{1+x^8}}{=} \frac{1}{(1+x^8)^2} \int_0^{+\infty} z e^{-z} dz = \frac{1}{(1+x^8)^2},$$

因此

$$\begin{aligned} \int_0^{+\infty} \frac{1}{(1+x^8)^2} dx &= \int_0^{+\infty} \left(\int_0^{+\infty} y e^{-(1+x^8)y} dy \right) dx = \int_0^{+\infty} \left(\int_0^{+\infty} y e^{-(1+x^8)y} dx \right) dy \\ &= \int_0^{+\infty} y e^{-y} \left(\int_0^{+\infty} e^{-x^8 y} dx \right) dy \stackrel{x=y^{-\frac{1}{8}} z^{\frac{1}{8}}}{=} \int_0^{+\infty} y e^{-y} \left(\int_0^{+\infty} y^{-\frac{1}{8}} e^{-z} dz^{\frac{1}{8}} \right) dy \\ &= \frac{1}{8} \int_0^{+\infty} y^{\frac{7}{8}} e^{-y} \left(\int_0^{+\infty} z^{-\frac{7}{8}} e^{-z} dz \right) dy = \frac{1}{8} \int_0^{+\infty} y^{\frac{7}{8}} e^{-y} \Gamma\left(\frac{1}{8}\right) dy \\ &= \frac{1}{8} \Gamma\left(\frac{15}{8}\right) \Gamma\left(\frac{1}{8}\right) = \frac{1}{8} \cdot \frac{7}{8} \Gamma\left(\frac{7}{8}\right) \Gamma\left(\frac{1}{8}\right) \\ &\stackrel{\text{余元公式}}{=} \frac{7\pi}{64 \sin\left(\frac{\pi}{8}\right)} = \frac{7\pi}{32\sqrt{2-\sqrt{2}}}. \end{aligned}$$

□

例题 0.21 计算积分 $I = \int_{-1}^2 \frac{1+x^2}{1+x^4} dx$.

注 在此例中 $I \neq F(2) - F(-1)$. 这是因为 F 并不是 f 在区间 $[-1, 2]$ 上的原函数.

解 Show/Hide Solution

在不包含 0 的区间上作变换 $t = x - \frac{1}{x}$ 得

$$\begin{aligned} \int \frac{1+x^2}{1+x^4} dx &= \int \frac{x - \frac{1}{x}}{2 + \left(x - \frac{1}{x}\right)^2} dx = \int \frac{dt}{2+t^2} \\ &= \frac{1}{\sqrt{2}} \arctan \frac{t}{\sqrt{2}} + C = \frac{1}{\sqrt{2}} \arctan \frac{x^2 - 1}{\sqrt{2}x} + C. \end{aligned}$$

这说明在区间 $[-1, 0)$ 和 $(0, 2]$ 上, 函数 $f(x) = \frac{1+x^2}{1+x^4}$ 的一个原函数是

$$F(x) = \frac{1}{\sqrt{2}} \arctan \frac{x^2 - 1}{\sqrt{2}x}.$$

因此

$$\begin{aligned} \int_{-1}^0 f(x) dx &= F(0^-) - F(-1) = \frac{\pi}{2\sqrt{2}} - 0 = \frac{\pi}{2\sqrt{2}}, \\ \int_0^2 f(x) dx &= F(2) - F(0^+) = \frac{1}{\sqrt{2}} \arctan \frac{3}{2\sqrt{2}} + \frac{\pi}{2\sqrt{2}}. \end{aligned}$$

故

$$I = \int_{-1}^0 f(x) dx + \int_0^2 f(x) dx = \frac{\pi}{\sqrt{2}} + \frac{1}{\sqrt{2}} \arctan \frac{3}{2\sqrt{2}}.$$

□

例题 0.22 计算 $I(b) = \int_0^{\frac{\pi}{2}} \frac{1}{b + \tan x} dx$.

解 Show/Hide Solution

解法一: 注意到

$$I(b) = \int_0^{\frac{\pi}{2}} \frac{1}{b + \frac{\sin x}{\cos x}} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\sin x + b \cos x} dx,$$

记

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\sin x + b \cos x} dx, \quad J = \int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + b \cos x} dx.$$

从而

$$bI + J = \int_0^{\frac{\pi}{2}} \frac{\sin x + b \cos x}{\sin x + b \cos x} dx = \frac{\pi}{2},$$

$$I - bJ = \int_0^{\frac{\pi}{2}} \frac{\cos x - b \sin x}{\sin x + b \cos x} dx = \ln(\sin x + b \cos x) \Big|_0^{\frac{\pi}{2}} = -\ln b.$$

联立上面两式解得

$$I(b) = I = \frac{\pi b}{2(b^2 + 1)} - \frac{\ln b}{b^2 + 1}.$$

解法二: 令 $t = \tan x$, 则

$$I(b) = \int_0^{\frac{\pi}{2}} \frac{1}{b + \tan x} dx = \int_0^{+\infty} \frac{1}{(b+t)(1+t^2)} dt = \lim_{A \rightarrow +\infty} \int_0^A \frac{1}{(b+t)(1+t^2)} dt.$$

注意到

$$\begin{aligned} \int_0^A \frac{1}{(b+t)(1+t^2)} dt &= \frac{1}{b^2+1} \int_0^A \frac{1}{b+t} dt + \frac{1}{b^2+1} \int_0^A \frac{-t+b}{1+t^2} dt \\ &= \frac{1}{b^2+1} \ln \frac{b+A}{b} - \frac{1}{2(b^2+1)} \int_0^A \frac{2t-2b}{1+t^2} dt \\ &= \frac{1}{b^2+1} \ln \frac{b+A}{b} - \frac{1}{2(b^2+1)} \int_0^A \frac{2t}{1+t^2} dt + \frac{b}{b^2+1} \int_0^A \frac{1}{1+t^2} dt \\ &= \frac{1}{b^2+1} \ln \frac{b+A}{b} - \frac{1}{2(b^2+1)} \ln(1+A^2) + \frac{b}{b^2+1} \arctan A \\ &= \frac{1}{b^2+1} \ln \frac{b+A}{b\sqrt{1+A^2}} + \frac{b}{b^2+1} \arctan A, \end{aligned}$$

于是

$$\begin{aligned} I(b) &= \lim_{A \rightarrow +\infty} \int_0^A \frac{1}{(b+t)(1+t^2)} dt \\ &= \lim_{A \rightarrow +\infty} \left[\frac{1}{b^2+1} \ln \frac{b+A}{b\sqrt{1+A^2}} + \frac{b}{b^2+1} \arctan A \right] \\ &= \frac{\pi b}{2(b^2+1)} - \frac{\ln b}{b^2+1}. \end{aligned}$$

□